BRCA model details and manual calculations

A numerical teaching companion for medical students

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Where this model came from

We used records from 1,295 past patients whose BRCA test results were already recorded. The computer looked for combinations of age, family history and tumour features that helped distinguish positive results from other recorded results.

The calculator applies that learned pattern to the inputs you enter. We checked its performance within the original patient group, but we still need a separate group of patients to show how reliably it works beyond that group. Having more records in the archive does not automatically provide that proof.

Learn how the stored model was trained, what every mathematical symbol means, and how entered options lead to probability, card points and queue lanes. The examples are synthetic; they do not contain patient records. No model was refitted and no new biological analysis was performed.

Synthetic profile	Probability	Card	Rule A	Rule B
Baseline: age 50	2.5%	0	Normal	Normal
Site example: age 38	25.8%	6	Urgent	Urgent
Card/model disagreement	8.6%	5	Urgent	Normal

The logistic probability, separately fitted points card, and A/B rules are three different layers. Six card points are not converted into 25.8%. The carrier probability applies to the historical cancer referral setting, not future cancer risk in healthy people. This teaching page does not grant students authority to make clinical decisions.

Open this English teaching page

Open the Turkish teaching page

How to use the companion

Read the training and limitations first, then the coefficient tables and symbol glossary. Follow the two-page worked example before changing a single option on the website. The final tables show how selected options and historical aggregate counts behave. The downloadable CSV contains all 54 synthetic examples.

Training data and model fitting

Development data and outcome

The original development group contains 1,295 people: 167 labelled Positive and 1,128 with other labels; 1,088 in the breast group and 207 ovarian-only, diagnosed in 1988–2020. The outcome y equals 1 only when BRCA_status_v3_strict is exactly Positive; other labels become 0. This does not establish complete, up-to-date clinical negativity for every zero. Neither the expanded archive nor the unvalidated 87-family tumour-size finding has been added to the model.

Preprocessing

Family counts were converted to numbers and missing values filled with zero. Missing age used the diagnosis-group median plus a missingness indicator. Missing region maps to Marmara; grades other than 1 or 3 map to the g2 reference. Missing Ki-67 uses 30 plus a separate indicator set to 1. The panel gives no personal result for blank age; ovarian-only selection clears TNBC and sets Ki-67 to missing.

Model fitting

Class-balanced L2 logistic regression used max_iter = 5000 and seed = 2020. Candidate C values were 0.01, 0.03, 0.1, 0.3 and 1. The main script used 5 outer folds × 5 repeats, with 3 inner folds selecting C by AUC. The modal selected C was 0.3; the final model was fitted to the complete development data. Age, family counts and Ki-67 were standardised for fitting; exported coefficients were converted back to raw units. The live panel does not standardise them again.

Recorded development field	Count
People	1295
Positive-labelled	167
Other labels	1128
Breast group	1088
Ovarian only	207
Age missing	87
Region unrecorded	16
Ki-67 recorded	464 people; 51 positive

Diagnosis breakdown (people / positives): unilateral breast 978/104; bilateral breast 75/17; breast and ovarian 22/10; male breast 13/2; ovarian only 207/34. The first four groups sum to 1,088. These counts are not the expanded archive's 2,148 operational identities or the 87-family tumour-size comparison.

Calibration validation and limitations

Calibration and the separate queue evaluation

The main script averages repeated out-of-fold raw probabilities for each person and fits Platt calibration to their logits. The panel applies the stored a and b to the final-model z score. The combined_policy script instead learns scaling on outer training data, calibrates using inner out-of-fold predictions, and counts queue outcomes in the outer test fold. These are distinct evaluation pipelines.

Validation limitations

The recovered main script imputes medians and standardises before the outer split. The queue script learns outer-fold scaling on training data but still uses pre-imputed data. StratifiedKFold does not group families. Therefore neither fully fold-contained preprocessing nor independence of relatives is guaranteed. The AUC interval uses 2,000 person-level bootstrap samples; optimism correction uses 200 bootstrap samples. These are not independent clinical validation.

Balanced class weights on the full development data

$$w_{+} = \frac{1295}{2 \times 167} = 3.877246\,,\; w_{-} = \frac{1295}{2 \times 1128} = 0.574025$$

Each class has total weight 647.5. Fold-level weights use that fold's training counts. Weighting does not add patients and is one reason the raw sigmoid output must not be treated as the final clinical frequency.

Stored performance is not independent validation

The package records repeated out-of-fold raw-score AUC 0.77647 (person-bootstrap interval 0.73285–0.81740), apparent AUC 0.81541, optimism-corrected AUC 0.78391, and temporal AUC 0.78837. The temporal split used 796 people/112 positives for training and 477/52 for testing. A monotone calibration map does not change score ranking; calibration quality and discrimination are different questions.

The recovered training script's old opening comment mentions 1,359/271; final package counts are 1,295/207. The final package is used here. Additional nested_validation fields have not had their complete generating chain reconstructed. The original scripts were read, not executed for this teaching companion.

All raw unit numerical coefficients

Use raw years, percentages and relative counts. Do not standardise again. The intercept is separate from the 26 coefficient terms.

$$z = \beta_0 + \sum_{j=1}^{26} \beta_j x_j$$

Term	Coefficient β
Intercept	-1.6082835151
Age at diagnosis	-0.0297415193
First-degree breast cancer <40	1.6177740836
First-degree breast cancer ≥40	0.6753591496
Second-degree breast cancer <40	0.6125414200
Second-degree breast cancer ≥40	0.5663237044
Third/fourth-degree breast cancer	0.3860573103
First-degree ovarian cancer	0.8656234065
Second–fourth-degree ovarian cancer	0.7417279306
First-degree other cancer	0.2127431762
Age missing indicator	1.0670551461
TNBC indicator	0.8095037558
Ki-67 (%)	0.0250358218
Ki-67 missing indicator	0.3771726605

Missing Ki-67 uses 30 plus its indicator. Blank age in the app produces no personal result despite the stored age_missing coefficient. Reference categories are breast, g2 and marmara.

Categorical coefficients and equivalent calibration

Selected category	Coefficient β
breast / g2 / marmara	0 (references)
dx=bilateral	0.8178367696
dx=breast_ovary	1.1847477030
dx=male	0.4984724193
dx=ovarian	0.3171356592
grade=g1	0.0897859062
grade=g3	0.6437786005
region=black_sea	-0.0922774437
region=eastern_anatolia	-0.2153762678
region=middle_anatolia	0.1604512937
region=balkans	0.4053551943
region=southeastern_anatolia	-0.3842514179
region=aegean	0.3473754169
region=mediterranean	-0.6314707319

$$\eta = a + b\,z\,,\; p_{cal} = \frac{1}{1 + e^{-\eta}}$$

$$a = -1.8240957468955834\,,\; b = 0.939227549587539$$

$$\beta_j^{*} = b\,\beta_j\,,\; \beta_0^{*} = a + b\,\beta_0 = -3.334639931778$$

The starred coefficients provide an equivalent one-pass calculation of η. Do not apply Platt calibration a second time. A category coefficient is conditional on the other recorded inputs; it is not evidence of a causal effect of region, grade or diagnosis. Region is not an ancestry assay.

Symbol glossary

Symbol	Meaning
β_0	Intercept: the starting value for every patient's z score, −1.608283515050.
β_j	Learned raw-unit coefficient for input j. It is not a points-card weight.
x_j	The patient's input j: years, percentage, relative count, or a 0/1 indicator.
j; 26	j indexes the inputs. There are 26 coefficient terms, excluding the intercept.
Σ	Sum: add every $\beta_j \times x_j$ product from j = 1 to 26.
z	Raw linear score: intercept plus all weighted inputs. It is not a percentage.
p; p_ham / p_raw	p denotes a probability between 0 and 1. p_raw (Turkish: p_ham) is the sigmoid result before calibration.
η; p_son / p_cal	η (eta) is the calibrated linear score. p_cal (Turkish: p_son) is converted to the displayed percentage.
a; b	Platt calibration intercept and slope: a = −1.824095746896; b = 0.939227549588.
e; exp(t)	e ≈ 2.71828 is Euler's number. exp(t) means e raised to power t; t is the exponent. exp(−z) = e^{-z}.
log; logit(p)	log is the natural logarithm. logit(p) = log[p/(1−p)], converting probability to the linear-score scale.
β_j^*; β_0^*	The star labels equivalent calibrated coefficients: $\beta_j^* = b\beta_j$; $\beta_0^* = a + b\beta_0$. Do not calibrate again.
OR; odds	Odds = p/(1−p). OR is a ratio of odds, not a percentage-point difference or a causal effect.
c_model; c_missing; c_{30}	c_model is an input's contribution to z. c_missing and c_{30} are the contributions of missing Ki-67 and measured Ki-67 of 30%.
n; min(n,2)	Number of relevant relatives. min selects the smaller number; a card row counts at most two relatives.
S_card; s_card	Uppercase S is the total card score. Lowercase s is its lane code: 0, 1 or 2. Turkish labels use kart instead of card.
s_model	Probability lane: p <0.10 → 0; 0.10 ≤ p <0.20 → 1; p ≥0.20 → 2.
u	Urgency lane: 2 if a treatment decision awaits the result; otherwise 1 if the familial P/LP box is checked; otherwise 0.

Symbol glossary continued

Symbol	Meaning
A; B; max	A and B are decision rules. max selects the highest lane. Codes: 0 normal, 1 priority, 2 urgent.
min; round	min selects the smallest value. round selects the nearest integer; exact halves go to the even integer in the code.
N	Approximate “1 in N” description: round(1/p). This is not an observed patient count.
i; q_i; q_{i+1}	i is the zero-based index of the lower percentile knot; q_i and q_{i+1} are the probabilities at two adjacent knots.
Position; Fill	Position is the rank among development scores. Fill is the visual bar percentage. Neither is the patient’s probability.
π	Development-group positive proportion, 167/1295. This is not the geometric constant π ≈3.14.
w_+; w_-	Training weights assigned to the positive and other outcome classes. The +/− subscripts identify classes.
ΔN; ΔT	Delta means difference. In capacity calculations, ΔN is the package-count difference and ΔT the target-positive difference. This N is separate from “1 in N”.
TP; FN	True positives: carriers above the threshold. False negatives: carriers below the threshold.
TN; FP	True negatives: noncarriers below the threshold. False positives: noncarriers above the threshold.
NPV	Negative Predictive Value: noncarrier proportion below the threshold, TN/(TN+FN).
AUC; ROC	Area Under the Curve: area under the Receiver Operating Characteristic curve, plotting sensitivity against false-positive rate. A single threshold table cannot give AUC.
C; L2	C is the inverse regularisation-strength setting; smaller C gives stronger penalisation. L2 penalisation is based on squared coefficients.
y; 0/1	Training outcome y is 1 for the historical Positive label, 0 for other labels. Input indicators use 1 for present/selected and 0 for absent/not selected.
=; ≈; ×; /	= means equal; ≈ approximately; × multiplication; / or a fraction bar division. Minus denotes a negative value or subtraction.
<; ≤; >; ≥	Less than; less than or equal; greater than; greater than or equal. Equality at a threshold matters.
%; subscript; exponent	Multiply probability by 100 to obtain percent. Subscripts name terms; raw/cal are not multipliers. A superscript gives a power; the star on β* is a label, not a power.

Numbers use decimal points in this English companion. Commas are decimal separators in the Turkish version. Keep full precision while calculating and round only for display.

Worked example inputs and weighted sum

Synthetic profile: diagnosis age 38; unilateral breast cancer; grade 2 / unknown; Marmara / unknown; TNBC present; Ki-67 unknown; one first-degree relative with breast cancer diagnosed before age 40. All other family counts are known zero; both urgency boxes are off. This is the site's Example patient profile.

Step 1 Encode the inputs

Age = 38, TNBC = 1, early breast relative = 1. Unknown Ki-67 produces Ki-67 = 30 and ki67_missing = 1. Known age produces age_missing = 0. Reference categories contribute zero. The other 21 coefficient terms contribute zero.

Step 2 Multiply and accumulate

Term	Substitution	Contribution	Running total
Intercept	1 × (-1.6082835151)	-1.6082835151	-1.6082835151
Age at diagnosis	38 × (-0.0297415193)	-1.1301777338	-2.7384612488
First-degree breast cancer <40	1 × (1.6177740836)	1.6177740836	-1.1206871653
TNBC indicator	1 × (0.8095037558)	0.8095037558	-0.3111834094
Ki-67 (%)	30 × (0.0250358218)	0.7510746533	0.4398912439
Ki-67 missing indicator	1 × (0.3771726605)	0.3771726605	0.8170639044

$$z = -1.608283515050 + (38 \times -0.029741519309)$$
$$+ (1 \times 1.617774083556) + (1 \times 0.809503755804)$$
$$+ (30 \times 0.025035821777) + (1 \times 0.377172660529)$$
$$z = 0.817063904400$$

Step 3 Calculate the raw sigmoid probability

$$p_{raw} = \frac{1}{1 + e^{-0.817063904400}} = \frac{1}{1 + 0.441726704} = 0.693612733313$$

The raw probability is about 69.36%. It is not the displayed final probability: calibration follows on the next page.

Worked example calibration percentage and lanes

Step 4 Substitute z into the stored calibration

$$\eta = -1.824095746896 + \left(0.939227549588 \times 0.817063904400\right)$$

$$\eta = -1.824095746896 + 0.767408928786 = -1.056686818109$$

a and b were learned previously, not re-estimated for this patient. A negative η gives a final probability below 50%.

Step 5 Calculate the displayed percentage

$$p_{cal} = \frac{1}{1 + e^{-(-1.056686818109)}} = \frac{1}{1 + e^{1.056686818109}}$$

$$p_{cal} = \frac{1}{1 + 2.876823742} = 0.257943116995$$

$$100\, p_{cal} = 25.7943116995\,\% \approx 25.8\,\%$$

Apply calibration once. The final percentage is a carrier-probability estimate in the model's historical referral setting, not a future cancer-risk estimate.

Step 6 Calculate the separate card and rule

Age <40 gives 1 point; TNBC gives 2; the early first-degree breast relative gives 3. Missing Ki-67 adds a model contribution but gives no high-Ki-67 card point.

$$S_{card} = 1 + 2 + 3 = 6 \Longrightarrow s_{card} = 2$$

$$p_{cal} \geq 0.20 \Longrightarrow s_{model} = 2, u = 0$$

$$A = max\left(2, 2, 0\right) = 2, B = max\left(2, 0\right) = 2$$

Both lanes are urgent. A triggers the rapid package because the final lane is urgent; B triggers it because the model probability is ≥20%. Six points are not converted into 25.8%. B remains a proposed policy.

$$N = round\left(\frac{1}{0.257943116995}\right) = round\left(3.876824\right) = 4$$

The "about 1 in 4" phrase is a rounded description, not a count of four real patients.

Changing age tumour information and family counts

Every row below changes one option from the baseline: age 50, breast, g2, marmara, no TNBC, missing Ki-67, all family counts zero and no urgency. Rows are not cumulative.

Single change	Probability	Card	A lane
Age 38	3.4%	1	Normal
Age 39	3.3%	1	Normal
Age 40	3.3%	0	Normal
Age 50	2.5%	0	Normal
Age 62	1.8%	0	Normal
Age 70	1.4%	0	Normal
TNBC present	5.2%	2	Normal
Measured Ki-67 0%	0.9%	0	Normal
Measured Ki-67 20%	1.4%	0	Normal
Measured Ki-67 29%	1.7%	0	Normal
Measured Ki-67 30%	1.8%	1	Normal
Measured Ki-67 100%	8.5%	1	Normal
First-degree breast cancer <40 +1	10.4%	3	Priority
First-degree breast cancer ≥40 +1	4.6%	1	Normal
Second-degree breast cancer <40 +1	4.3%	1	Normal
Second-degree breast cancer ≥40 +1	4.1%	1	Normal
Third/fourth-degree breast cancer +1	3.5%	1	Normal
First-degree ovarian cancer +1	5.4%	1	Normal
Second–fourth-degree ovarian cancer +1	4.9%	1	Normal
First-degree other cancer +1	3.0%	0	Normal

A measured Ki-67 of 30% gives a lower model probability than missing Ki-67 in the baseline, but adds one card point. The missingness contribution is algebraically equivalent to measured Ki-67 ≈45.0653% in this model, not a biological imputation or a known patient value.

$$c_{missing} = 30 \times 0.025035821777 + 0.377172660529 = 1.128247313849$$

$$c_{30} = 30 \times 0.025035821777 = 0.751074653320$$

Categories family caps and clinical urgency

Single change from baseline	Probability	Card
dx = breast	2.5%	0
dx = bilateral	5.2%	2
dx = breast_ovary	7.2%	2
dx = male	3.9%	1
dx = ovarian	3.3%	2
grade = g2	2.5%	0
grade = g1	2.7%	0
grade = g3	4.4%	1
region = marmara	2.5%	0
region = black_sea	2.3%	0
region = eastern_anatolia	2.0%	0
region = middle_anatolia	2.9%	0
region = balkans	3.6%	0
region = southeastern_anatolia	1.7%	0
region = aegean	3.4%	0
region = mediterranean	1.4%	0

The separate card and decision rules

The card comes from a separate class-balanced L2 model with 16 candidate features and C = 0.3. Its coefficients were divided by 0.5 log-odds units and rounded; zero rows were removed, leaving 14 rows. It is not obtained by rounding the main model's β coefficients. Card lanes: 0–3 normal, 4 priority, ≥5 urgent. Model lanes: <10% normal, 10–<20% priority, ≥20% urgent. A takes the maximum of card, model and urgency; B takes model and urgency only. B remains a proposal. Rapid-package triggers: urgent final lane for A, model ≥20% for B.

Missingness and the familial-variant box

Unknown Ki-67 differs from measured 30%: the missingness indicator adds a model contribution but gives no high-Ki-67 card point. Each family card row is capped at two relatives; the model uses counts up to six. The familial P/LP box does not change the β sum, but the personal probability headline is suppressed because the model does not include this familial-variant information. Unknown information is not a known zero.

$$c_{model} = n \times 1.617774083556, S_{row} = 3\,min(n, 2)$$

Early first-degree counts 0/1/2/3/6 give probabilities 2.5/10.4/34.7/70.8/99.6% and card contributions 0/3/6/6/6. These are synthetic outputs. Treatment sets u = 2; familial P/LP alone sets u = 1. Neither alters z.

Rank graphics and hypothetical test reduction

Percentile position

Stored probability knots correspond to 0, 5, …, 100 percentiles. For the worked example, the lower knot has zero-based index 17 (85th percentile), q_{17} = 0.2133526175 and q_{18} = 0.2763791363.

$$Position = round\left[5\left(17 + \frac{0.2579431170 - 0.2133526175}{0.2763791363 - 0.2133526175}\right)\right] = 89$$

This is not an 89% carrier probability. It is an interpolated position among development scores. Only reaching the last stored knot returns 100 and activates the app's "top 5%" wording; do not reinterpret this as a separately validated percentile definition.

Visual probability bar

$$Fill = min\left(100, \frac{p}{0.40} \times 100\right)$$

At or above 40% probability the bar is full. A full bar does not mean 100% probability; read the percentage headline.

Research reduction scenario

The capacity panel reads aggregate outcomes for six stored strategies; it does not predict new patients. Cross-validation counts are repeat means; temporal counts concern 477 people and 52 carriers. The hypothetical 5% test-reduction line also excludes card ≥4, ovarian cancer, male breast cancer, TNBC, ovarian family history, urgency and, when enabled, age <45. Full testing remains the plan; this is not an instruction to omit testing.

At the baseline, p = 0.024803319313 and card = 0 satisfy the default hypothetical policy with age protection enabled. The 38-year-old worked example fails several criteria. The absent generating script and policy approval record mean the stored test-reduction results are exploratory and were not independently regenerated here.

What option changes do not do

Language changes and presentation navigation do not change model coefficients. Family counts of zero are a synthetic known-zero assumption in these examples; in actual records unknown is not zero. Age is continuous in the model, while the card gives an abrupt point change at age 40. A ten-year increase multiplies calibrated odds by 0.756282, conditional on fixed other inputs; it is not a causal ageing effect.

Arithmetic behind stored capacity and validation totals

Cross-validation model ≥20%: 212.6 packages and 26.6 detectable targets. Rule A: 280 packages and 29.4 targets. These are stored repeat means, not predictions for a new month.

$$\Delta N = 212.6 - 280 = -67.4\,,\ \Delta T = 26.6 - 29.4 = -2.8$$

$$Yield = \frac{26.6}{212.6} \times 100 = 12.511759\,\% \approx 12.5\,\%$$

$$Packages/target = \frac{212.6}{26.6} = 7.992481 \approx 8.0$$

The stored mean of repeat-specific ratios is 8.0129973; the screen uses the ratio of displayed means, 7.9924812. These are different operations. A 220-package cap admits 212.6 but excludes 227 or 280. Temporal data: model ≥20% selects 40 packages/8 targets; A selects 49/8.

The target denominator is 41 carriers of the rapid package's targets, not all 167 BRCA carriers. At model ≥20%, 41−26.6 = 14.4 target carriers remain outside rapid selection and stay on the full-test plan. No time, cost or surgical benefit follows from this arithmetic alone.

Historical 95 percent sensitivity threshold

At stored threshold 0.035019185420: 8 of 167 carriers fall below, and 158 people in total fall below. Thus TP = 159, FN = 8, TN = 150, FP = 978. This is not the 10%/20% queue policy or the 5%-plus-exclusions scenario.

$$Sensitivity = \frac{159}{167} = 95.2096\,\%\,,\ NPV = \frac{150}{158} = 94.9367\,\%$$

$$Below\ threshold = \frac{158}{1295} = 12.2008\,\%$$

$$\pi(1-\pi) = \frac{167}{1295} \times \frac{1128}{1295} = 0.112327484683$$

The last expression is the Brier baseline from assigning everyone the development prevalence. AUC requires score ordering or an ROC curve, not one 2×2 table.

Source provenance and what has actually been checked

The teaching calculations use the frozen aggregate model file and original training-script audit. The 54 synthetic profiles match the existing calculation core; 47 stored synthetic self-test vectors check core probabilities. This verifies arithmetic, not independent clinical validity. No patient data were reanalysed for these documents.

Exact bit-level reconstruction of training would additionally require frozen training-data versions, all supplementary generating scripts, software versions and fold assignments. The complete generating chain for additional nested_validation fields has not been reconstructed. Family grouping and preprocessing limitations remain material.

Current calculator source and frozen model package

Original training script in project Drive

Separate points-card training script in project Drive

Combined queue-evaluation script in project Drive

Drive source links require existing project access. The website provides the aggregate coefficient table, the synthetic-example CSV and this PDF without granting access to individual patient folders. The research report retains its separate date of 25 October 2026; this student guide is dated 2 November 2026.

Abbreviations used in this companion

BRCA1/BRCA2: BRCA1/BRCA2 DNA repair associated genes. TNBC: Triple-Negative Breast Cancer (ER−, PR−, HER2−). ER: Estrogen Receptor; PR: Progesterone Receptor; HER2: Human Epidermal Growth Factor Receptor 2. Ki-67: proliferation-marker percentage. MLPA: Multiplex Ligation-dependent Probe Amplification, used to assess deletions/duplications. P/LP: Pathogenic/Likely Pathogenic; VUS: Variant of Uncertain Significance. AUC, ROC, NPV and OR are explained in the symbol glossary. JSON and CSV are structured data-file formats; they are not clinical measurements.